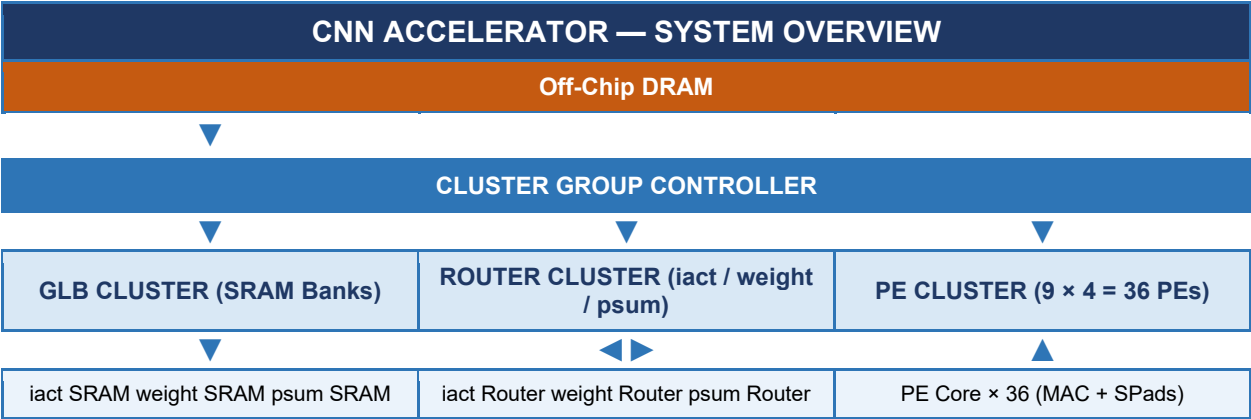


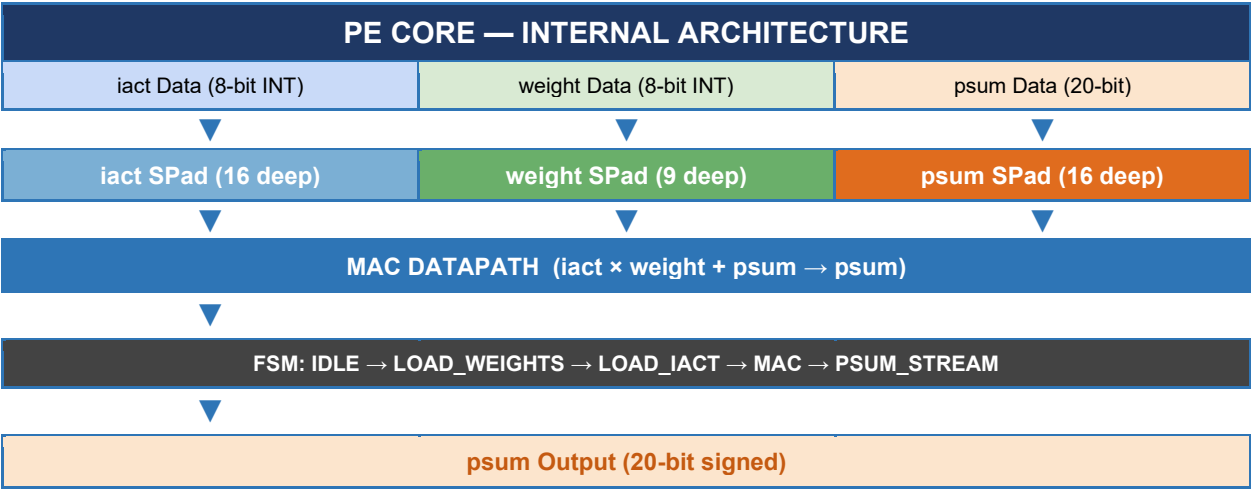
Eyeriss-v2 CNN Accelerator — RTL Architecture Diagrams

Team SMC26-30 | Target: LEGNet on FPGA | Dataflow: Hierarchical Mesh NoC + CSC Encoding

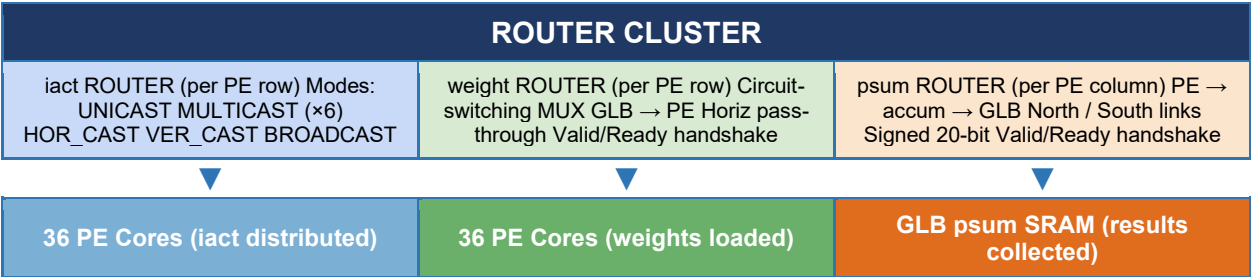
1. System-Level Overview



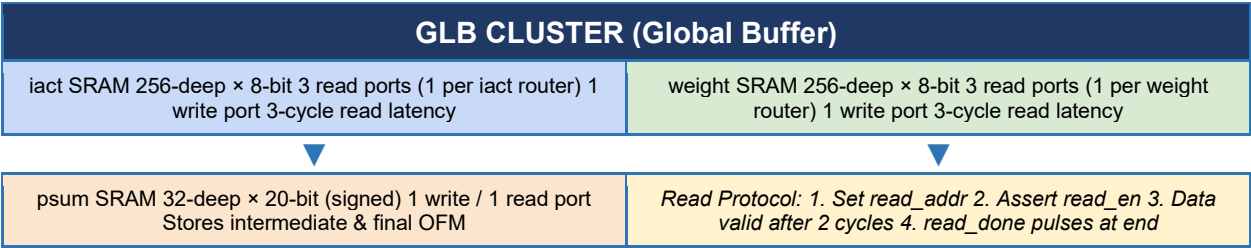
2. Processing Element (PE) Core



3. Router Cluster



4. Global Buffer (GLB) Cluster



5. Convolution Dataflow

DATAFLOW — IACT-STATIONARY / WEIGHT-STATIONARY HYBRID			
Step	Action	Data Path	Data Type
1	Load weights from GLB	GLB weight SRAM → weight router → weight SPad	8-bit INT (stationary)
2	Stream iacts with multicast reuse	GLB iact SRAM → iact router → iact SPad (shared across rows)	8-bit INT (multicast)
3	MAC operations in each PE	iact SPad × weight SPad → accumulated into psum SPad	20-bit signed accumulator
4	Stream psums to GLB	psum SPad → psum router (accumulated per column) → GLB psum SRAM	20-bit signed output

All signals use valid/ready handshake. Precision: 8-bit INT (iact & weight), 20-bit signed (psum). PE array: 9 rows × 4 cols = 36 PEs per cluster.